LAVA ESCAPE

Final Project Submission Report

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Course: B.Tech — Computer Science & Engineering (AI & ML)

GitHub Repository	https://github.com/hemantchalavadi182-sudo/52_lava_escape
Project Type	Python / Pygame platform game
Development Environment	Ubuntu (WSL) with Python 3 and Pygame

This PDF is a submission report and guide to the project. The complete runnable source files, Git history, and any uploaded gameplay recordings are maintained in the GitHub repository linked above. A PDF is not a replacement for the original Python files or video files.

1. Project Overview

Lava Escape is a vertical platform game in which the player jumps between platforms while avoiding rising lava. This project extends the starter game with improved platform collision behavior, temporary hazards, spring-assisted jumps, and a danger indicator with periodic lava surges.

2. Implemented Tasks

Task 1 — One-way platform collision

Corrected platform landing behavior so the player lands when descending across a platform's top edge. The player can pass through a platform from below instead of being incorrectly blocked by its underside.

Task 2 — Crumbling platforms

Added temporary platform hazards that begin shaking after the player lands and disappear after a short delay.

Task 3 — Spring platforms

Added special platforms that launch the player upward with increased velocity.

Task 4 — Danger HUD and lava surges

Added a danger display and periodic surge phases that temporarily increase the lava's rise speed.

3. Controls

Action	Controls
Move left / right	A / D or Left / Right arrow keys
Jump	Space, W, or Up arrow
Restart	R

4. Git Commit History

C	D
0	F
5	A
0	A
d	A
c	A
P	B

5. Project Files and Evidence

Use the GitHub repository as the source of truth for all runnable project files. The repository is the correct place to submit Python source code, README instructions, and the recorded gameplay evidence.

Expected video evidence

P	P
v	A
v	A

Evidence note: Confirm that both files exist in the repository's *videos* folder and that *before-video.mp4* genuinely shows the original game. The conversation confirms the after video was pushed in commit c031784; the final before-video replacement's push status has not been confirmed in the available record.

6. Running the Game

1. Clone or download the repository from GitHub.
2. Open a terminal in the project directory.
3. Create and activate a Python virtual environment (recommended).
4. Install Pygame if it is not already installed.
5. Run the project's main Python entry point as documented in the repository README.

Example setup commands

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install pygame
```

The exact launch command should be taken from the repository README, since the entry-point filename is not verified in this report.

7. Final Submission Checklist

■ Open the GitHub repository and confirm the latest code is visible.

■ Confirm the four feature commits are present in the commit history.

■ Confirm videos/before-video.mp4 shows the original game and is pushed.

■ Confirm videos/after-video.mp4 shows the completed game and is pushed.

■ Review the README for setup instructions and evidence links, if required.

■ Submit both the repository link and this PDF if your instructor requests a report.

■ If required by the lab instructions, separately provide the ChatGPT conversation share link.

Repository link

https://github.com/hemantchalavadi182-sudo/52_lava_escape

Prepared on 28 October 2026. This report summarizes the project state described during the working session; the GitHub repository should be checked before final submission.